Experiment No: 01

Title: Implementation of DDL commands of SQL with suitable examples (Create table, Alter table, and Drop)

Topics to be covered:

- Create table
- Alter table
  - Alter table….ADD
  - Alter table…. MODIFY
  - Alter table…. DROP
  - Alter table…. RENAME
- Drop table
- Rename table

Introduction to SQL Data Types:

Categories of SQL Data Types

Category	Example Types
Character	CHAR, VARCHAR, TEXT
Numeric	INT, FLOAT, DECIMAL, NUMERIC
Date & Time	DATE, TIME, DATETIME, TIMESTAMP
Boolean	BOOLEAN, BIT
Binary	BINARY, VARBINARY, BLOB
Special/Other	ENUM, SET, JSON (in some DBMS)

Character/String Data Types

Type	Description	Syntax Example
CHAR(n)	Fixed-length string (up to 255 chars). Pads with spaces.	CHAR(10)
VARCHAR(n)	Variable-length string (up to 65535 bytes).	VARCHAR(50)
TEXT	Large text (up to 65,535 chars).	TEXT

Use CHAR for fixed-length codes (e.g., country codes), VARCHAR for names/descriptions.

Numeric Data Types:

Integer Types:

Type	Description	Range
TINYINT	1 byte integer	-128 to 127
SMALLINT	2 bytes	-32,768 to 32,767
INT	4 bytes	-2,147,483,648 to 2,147,483,647
BIGINT	8 bytes	Very large numbers

Decimal Types:

Type	Description	Example
DECIMAL(p,s)	Exact value. Good for currency.	DECIMAL(8,2) = 123456.78
NUMERIC(p,s)	Same as DECIMAL	
FLOAT	Approximate floating point number	FLOAT(7,4) = 123.4567
REAL/DOUBLE	Higher precision float (platform dependent)	

Date and Time Data Types:

Type	Description	Format
DATE	Stores only date	YYYY-MM-DD
TIME	Stores only time	HH:MM:SS
DATETIME	Stores date and time	YYYY-MM-DD HH:MM:SS
TIMESTAMP	Similar to DATETIME (auto-updated)	Depends on DBMS
YEAR	Stores a year	YYYY

Boolean and Binary Types:

Boolean:

Type	Description
BOOLEAN	Stores TRUE or FALSE (often as 1/0)
BIT	Can store bits (0 or 1)

Binary:

Type	Description
BINARY(n)	Fixed-length binary data
VARBINARY(n)	Variable-length binary data
BLOB	Binary Large Object (e.g., images)

Special Types (Optional/Advanced):

Type	Description
ENUM	A predefined list of string values
SET	A set of string values
JSON	Stores JSON-formatted data (in MySQL, PostgreSQL)

Creating a Database: To begin working with SQL, you first need to create a database.

Syntax:

```
CREATE DATABASE database_name;
```

Example:

```
CREATE DATABASE BAUET;
```

This command creates a new database named `BAUET`.

To view existing database names, the command is: **SHOW DATABASES ;**

<u>SQL DDL Commands (CREATE, ALTER, DROP, RENAME)</u>

1. CREATE TABLE: Used to create a new table (relation) in the database.

Syntax:

```
CREATE TABLE table_name (
    column1 data_type(size),
    column2 data_type(size),
    ...
);
```

Example:

// Simple format	// Standard format

`CREATE TABLE Student (`
`  sno int(3),`
`  sname CHAR(10),`
`  class CHAR(5)`
`);`
	`CREATE TABLE Student (`
	`  StudentID INT PRIMARY KEY,`
	`  FirstName VARCHAR(50) NOT NULL,`
	`  LastName VARCHAR(50),`
	`  Age INT CHECK (Age > 0),`
	`  EnrollmentDate DATE`
	`);`

2. ALTER TABLE: **Used to** modify the structure **of an existing table.**

(a) Add New Column(s): Adds one or more new columns to a table.

Syntax:

```
ALTER TABLE table_name
ADD (
    column1 data_type(size),
    column2 data_type(size),
    ...
);
```

Example:

```
ALTER TABLE Student
ADD (
    Address CHAR(10)
);
```

(b) Modify Existing Column(s): Changes the data type or size of one or more columns.

Syntax:

```
ALTER TABLE table_name
MODIFY (
    column1 new_data_type(size),
    column2 new_data_type(size),
    ...
);
```

Example:

```
ALTER TABLE Student
MODIFY sname VARCHAR(10),
MODIFY class VARCHAR(5);
```

Also it is possible by using RENAME

```
ALTER TABLE Student
CHANGE sname sname VARCHAR(10),
CHANGE class class VARCHAR(5);
```

(c) Drop a Column: Removes a column from a table.

Syntax:

```
ALTER TABLE table_name
DROP COLUMN column_name;
```

Example:

```
ALTER TABLE Student
DROP COLUMN sname;
```

(d) Rename a Column: Changes the name of an existing column.

Syntax1: Using the RENAME Statement

```
ALTER TABLE table_name
RENAME COLUMN old_column_name TO new_column_name;
```

Example:

```
ALTER TABLE Student
RENAME COLUMN sname TO stu_name;
```

Syntax2: Using the CHANGE Statement

```
ALTER TABLE table_name
CHANGE COLUMN old_column_name new_column_name datatype;
```

Example:

```
ALTER TABLE Student
CHANGE COLUMN sname stu_name VARCHAR(20);
```

3. RENAME TABLE: Renames an existing table to a new name.

Syntax:

```
RENAME TABLE old_table_name TO new_table_name;
```

Example:

```
RENAME TABLE Student TO StudentInfo;
```

4. DROP TABLE: Permanently deletes the entire table structure and all its data.

Syntax:

```
DROP TABLE table_name;
```

Example:

```
//DROP TABLE Student;


DROP TABLE StudentInfo;
```

Combined Example:

Below is a step-by-step implementation of all the commands:

```
-- Step 1: Create a database
CREATE DATABASE College;

-- Step 2: Switch to the new database
USE College;

-- Step 3: Create a table
CREATE TABLE Faculty (
    FacultyID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Department VARCHAR(50),
    JoiningDate DATE
);

-- Step 4: Alter the table to add a new column
ALTER TABLE Faculty
ADD Email VARCHAR(100);

-- Step 5: Rename the table
RENAME TABLE Faculty TO Professors;

-- Step 6: Drop the table
DROP TABLE Professors;
```

LAB Task-1:

1. **Create a table** called EMPLOYEE with the following columns:

- Emp_no (integer)
- E_name (varchar(100))
- E_address (varchar(200))
- E_ph_no (varchar(15))
- Dept_no (integer)
- Dept_name (varchar(100))
- Job_id (char(2))
- Salary (decimal(10, 2))

2. **Add a new column** HIREDATE (date) to the existing EMPLOYEE table.
3. **Modify the datatype** of the Job_id column in the EMPLOYEE table from CHAR to VARCHAR(2).
4. **Rename the column** Emp_no to E_no in the EMPLOYEE table.
5. **Change the column width** of the Job_id column in the EMPLOYEE table to accommodate a larger size.
6. **Drop the column** Dept_name from the EMPLOYEE table.
7. **Rename the EMPLOYEE table** to STAFF.
8. **Delete the EMPLOYEE table** from the database.

LAB Task-2:

1. Create a table EMPLOYEE with the following schema:

   (Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name, Job_id, Salary)
2. Add a new column; HIREDATE to the existing relation.
3. Change the datatype of JOB_ID from char to varchar.
4. Change the name of column/field Emp_no to E_no.
5. Modify the column width of the job field of the emp table
6. Drop the Dept_name column from the EMPLOYEE table
7. Rename the EMPLOYEE table to STAFF
8. Increase the salary of all employees by 10%
9. Remove the EMPLOYEE table from the database